For each question, show all related work and explain the solution.

1. (5 pts) Determine whether $168 \equiv 24 (mod 16)$

2. (15 pts) The octal expansion of an integer is $(4205)_8$. Write the decimal expansion, binary expansion and hexadecimal expansion.

3. (15 pts) Use the Euclidean algorithm to write $gcd(21, 55)$ as a linear combination of 21 and 55. Be sure to show all steps that lead to your solution.

4. (15 pts) Solve the congruence by first finding the inverse. $89x = 2\ (mod\ 232)$

5. (15 pts) Prove using mathematical induction that $\sum_{i=1}^{n}\left(\frac{1}{i(i+1)}\right) = \frac{n}{n+1}$, when n is a non-negative integer. Don't forget to begin with, "Let P(n) be the statement…".

6. (15 pts) Prove that $3^n < n!$, whenever n is an integer with $n \geq 7$.

7. (10 pts) Suppose that $\{a_n\}$ is defined recursively by $a_{n+1} = (a_n)^2 - 1$ and that $a_0 = 5$. Find $a_3$and a_4.

8. (10 pts) Give a recursive algorithm for computing na using **addition**, where n is a positive integer and a is a real number.